

CHAPTER 12

RACING IN THE RAIN

UNLESS YOU'RE WILLING to withdraw from a racing event when it rains—an act which carries with it a certain stigma of dishonor—sooner or later you'll have to come to terms with how rain affects your racing performance. Other than moving to Arizona, there are numerous adaptations that can be made to increase the chance of racing success when it turns crummy. It's worth spending some time planning how you're going to deal with rain before you're actually confronted with it on the track.

Car Control

There are two fundamental problems with racing in the rain: restricted visibility and loss of traction. Both problems have to be dealt with in order to retain control of your car in wet weather.

Obviously, first and foremost, you can't control your car if you can't see where you're going. Maintaining the ability to see comes even before car changes as a priority in wet driving.

Second, it's possible for your tires to lose all contact with the racetrack; and when they do, all your ability to control the car's speed and direction goes

There are three major areas in which you can gain the advantage by altering your dry-weather approach to racing:

- 1) maintaining the ability to see,**
- 2) altering the racecar, and**
- 3) rethinking driving technique.**

out the window. This happens when the water sitting on top of the asphalt is allowed to pack up between the tire and the road—you're literally driving on water, not asphalt. The phenomenon is

called "hydroplaning," and it can happen in your street car, in an airplane on the runway, in anything where a tire is rolling over a wet surface.

If you can handle those two problems, then you can go on to make the changes in driving technique that will be required to lower lap time in the wet.

MAINTAINING VISIBILITY

There are a few problems in maintaining visibility: some which you can solve and some you're stuck with. In the "stuck with" category, you have the spray off the wheels of the other cars in the field. In open-wheel cars, it's the worst, since every car is sending up a roostertail of water from each tire and the spray just hangs in the air. Other than being on the front row, there is not much you can do to improve the view, especially in the early laps when the field is bunched.



Fig. 12-1. Spray, especially off the wheels of an open-wheel racecar, makes visibility difficult for following cars.

"I think driving at the limit in the rain takes more thought than driving at the limit in the dry where conditions don't change drastically. In the dry, pretty much, the track is a constant. Sure, it changes, but it doesn't change every second. In the rain, what you can see changes constantly. The spray changes, it rains harder on one end of the track than the other—I think it's much more a thinking game.

There were times racing at Mosport in Canada where you would get wheelspin ten times a lap going down the straightaway. It's not like the dry where the drama is in the turns, then you're waiting for the next corner. You're driving it on the limit all the time."

—Skip Barber

A problem you *can* work on is to try to keep your visor (or goggles if you wear an open faced helmet) from fogging over. A full-face helmet on a rainy day is the ideal environment for mist to collect on the inside of the visor: the relative humidity is high, the temperature of the visor is colder than the air inside the helmet, and your heavy breathing is supplying an ample source of warm, humid air.

There are a number of potential fixes for the problem. One is to get outside air flowing over the inside of the visor. On most helmets you can ratchet the visor up a quarter inch or so, leaving a slot at the bottom wide enough for air to get through but narrow enough to keep water out. To keep the slot from closing down from wind pressure, you can roll up a quarter-inch cylinder of racer's tape and tape it between the helmet and visor so that the face shield can't close.

The ventilation normally helps, but you still have to deal with the warm, moist air you're breathing out. You can easily solve this problem by creating a cone of racer's tape, taped to your nose, that directs your breath away from the visor and down out the bottom of the helmet. Some drivers have gone as far as using a flexible snorkel or modified aircraft oxygen mask to get the job done. Whatever works!

Coating the Visor

There are preparations you can apply to the inside of the shield to cut down on fogging. "Fog Free" is a product you can buy at most racing or motorcycle shops and it works pretty well most of the time. Many drivers have had success coating the inside of the visor with liquid dishwashing detergent. Applied in a thin film of liquid, it doesn't distort the view and any condensation that forms turns instantly to a liquid you can see through. Lemony-fresh Joy is a special favorite since it also functions as an air freshener.

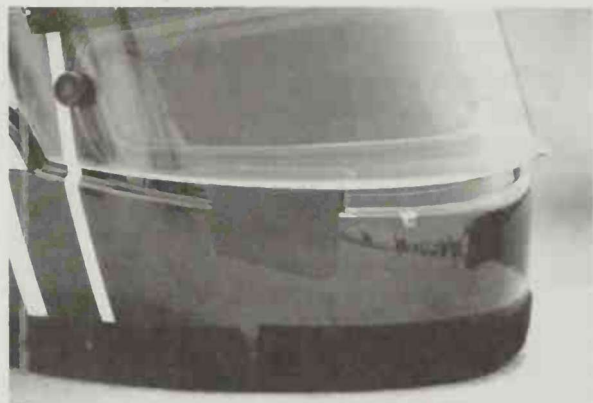


Fig. 12-2. To encourage air flow over the inside surface of your visor, prop the bottom edge of the visor open with a small cylinder of racer's tape.

For the outside of the visor, some drivers use "Rain-X," a formulation which beads water, but others don't like the idea of a bead forming directly in front of an eyeball. In open cars, the air rushing by normally blows the visor clear, and a good clean visor stays transparent without any special coating. Avoid using "tear-offs," thin transparent sheets attached to the visor that can be torn off as the race progresses. In the rain, water gets between the sheets and distorts the view.

"Believe it or not, somebody once told me to try tobacco on the inside of the visor to keep the fog down—and it works! You tear apart a cigarette and wipe the tobacco on the inside of the visor with your hand, then lightly wipe off the pieces with a cloth. I was amazed."

—David Loring

There has been a lot of experimentation with visor color in rain conditions. Obviously, you want to avoid using darkly-tinted visors in low-light conditions. A brand new, unscratched, clear visor gives you a real-world view. Yellow visors heighten contrast and many drivers like them. You do, however, have to check with the race-organizing body to see if they are allowable, as some groups feel that they make it difficult to distinguish the color of flags.

Mud and oily spray can be a problem and once the visor gets splattered, you can often do more harm than good in trying to clean the mess up. A useful trick is take a small towel, sponge or chamois along with you in the car and stow it under a shoulder belt or alongside your hip.

In closed cars, you need wipers in hard rain; and the inside of the windshield, as well as your visor, can

get foggy. Dorsey Schroeder and Walt Bohren, both IMSA GTO veterans, carry squeegees with them in rain races to wipe away the fog. If the squeegee has a long enough handle, you can use it to clean the inside of the windshield between gear shifts. You can use the same anti-fog preparations on the windshield as you do on your visor. Windshield ventilation also helps, so think twice before discarding the defroster from a sedan racecar. You will someday wish you had it back.

"It's amazing. The most important thing is the visibility through the windshield, and so few [GTS] cars have proper wipers or defoggers on them. Big budgets, but nobody's getting it right, you know? When was the last time you heard a guy get out of the car after a rain race and say 'The wiper worked flawlessly—the visibility was great!'"

—Jeremy Dale

OPTIMIZING TRACTION

Once you're sure that you'll be able to see where you're going, you can concentrate on optimizing the car for slippery conditions.

If you normally run slicks in your form of racing, the first hard rain-related decision will be whether to run rain tires or not. If the track is merely damp and the weather looks like it's improving, you may choose to start the race on slicks in the expectation that the track will dry and you'll only suffer in the early laps. Of course, it might rain harder and you'll have to stop in the pits once the race is underway to change to rains. It's a guessing game, and if you race for a while, you'll guess wrong as often as you'll guess right.

If you're undecided up until the last minute, it's a good idea to roll the car to the grid with two rain tires and two slicks mounted to the car, so that when you make your final, last-minute decision, you'll only have to change two tires, not four.

Rain Tires

Racing rain tires are designed to accomplish two goals: 1) to evacuate water from under the contact patch, and 2) to allow the rubber compound to reach a temperature where it can provide maximum grip given the cooling effects of the water.

Here's a good model for how the rain tire works: the first third of the contact patch evacuates the water toward the side of the tire, the second third squeegees the road surface dry, and the final third provides grip, using the same bonding and shearing mechanisms that the dry tire uses.

To take into account the cooling effects of running in water, the rubber compound of a rain tire is soft enough that it will reach the low 200-degree range in racing conditions. If the tire is run in the dry, the added friction and the lack of cooling water will quickly overheat it, changing the molecular structure of the rubber as well as blistering and chunking the tire.

If you start a race on rains and the track dries, you'll be forced to change back to slicks. In a race where the final four to five laps are run in drying conditions, you'll see drivers driving off-line through the puddles in an effort to cool the tires enough to keep them from coming apart so that they can make it to the finish without having to pit.

If you race in an area where it rains infrequently, you have to be careful about using old rain tires that have been sitting in the trailer for two seasons. Time slowly deteriorates the rubber compound, and you could find yourself at a distinct disadvantage running against other drivers with freshly-purchased rains.

"If we have different types of rain tires available, we'll test tires in the rain. It depends. If we have something specific we want to try in a rain situation, we'll go out, but not just to run around."

—Bryan Herta

Shaved Street Tires

In classes such as showroom stock where you normally run on shaved street tires, your rain choice should be the same type of tire with more tread thickness. How much more depends upon the depth of water involved. In a deluge, you'd opt for full tread depth, but with less and less water on the track, you can go progressively thinner on the tread. With our Formula Dodges, 5/32 inches of tread depth is about as deep as you can go without affecting dry track lap time. In light rain or in drying conditions, tires with 5/32 inches of tread will evacuate water well and not give away anything if the track dries. In order to know for sure what suits your racecar, you have to test different combinations.

One useful experiment for showroom stock would be to try to run a treaded tire with a higher aspect ratio: for example, a 70-series vs. a 60-series tire. The narrower contact patch would enhance braking and acceleration grip and sacrifice some cornering potential. Since cornering potential tends to suffer in the wet in greater proportion than accelerating and braking, boosting the car's existing strengths may be to your advantage.

We also recommend boosting inflation pressures as a potential fix for hydroplaning. Airline pilots have

known for years that the hydroplaning speed of a tire increases as the square root (multiplied by a constant) of the inflation pressure, and this remedy might be worth trying on your racecar.

CHANGING YOUR SET-UP

When it comes to modifying the mechanical set-up of the car for the rain, we can offer some suggestions but, from decades of experience, we've found that what works for some cars doesn't necessarily work for others. The only definitive way to find advantages for your car is to continually experiment and test every change you make.

"Most teams don't come up with a strictly wet set-up. The reason is that when it rains, changes in the weather are unpredictable; and if you go to a set-up that's good in the rain, generally it's absolutely horrible in the dry. With Indy Cars, they can dry the track so quickly that if it starts to lighten up and you get any dry at all, you're out to lunch with a wet-weather set-up. Conversely, a good dry set-up is not bad in the rain. You'll make some adjustments to the dry set-up, primarily to change the balance of the downforce. You can take front downforce out of the car to help secure the back. That's the only change you'd make because it doesn't totally ruin the car if the track dries."

—Bryan Herta

Testing in the wet is unpopular. You tow your perfectly clean and tidy racecar, fresh from its winter rebuild, to the track for testing and if it rains, it's the rare team who doesn't just turn around and go home. As distasteful as it might be to run in the slop, finding improvements in your wet-weather set-up will turn out to be a big asset by the end of the season, especially if your competition hasn't bothered to test in the wet.

Brake Bias Adjustments

It's almost impossible to say categorically that such and such a change will work for all cars. The one wet-weather tweak that comes close to being universal is changing the brake bias of the car: we've seen many a car do well in the rain without changing a thing, other than brake bias. With less grip in slippery conditions, there will be less deceleration and therefore less load transfer onto the front tires. If the brake bias was set in dry conditions, it will be proportioned too much to the front for the wet and will need to be adjusted toward the rear. To find out how much to adjust, you'll have to test as you go along.

"Testing in the wet is very important.

Penske was one of the only ones to test in the rain, but even there, they were reluctant. The problem with testing in the wet was that there's a big risk. You're worried that you could stick it in the fence or get hurt.

"We sat out at Portland one time for a couple of days—we were supposed to test for three days and it rained all three days. We sat around a lot until finally, on the third day, I said, 'You know what? It's wet. Let's go run.' We did because it wasn't puddling. If you've got puddles, you can really aquaplane and if you crash a car, you've just set yourself back so far. Even with a team like Penske. It takes about 800 hours, roughly, to build a car.

"The big thing to me was that we did it and developed a rain set-up that stayed with us for two years. Unless you're in a series that doesn't run in the wet, sometime you're going to have to do it."

—Danny Sullivan

Suspension Changes

Generally, we have seen success in going softer with shocks, springs and anti-roll bars in wet conditions. There are two reasons for this theory. First, when traction is limited, the driver can more easily deal with finding the limit when loads transfer gradually rather than suddenly. If the car has more motion in roll and pitch, which it would with softer suspension settings, individual tire loadings change more gradually, giving the driver a better chance at tiptoeing up toward the limit without overstepping.

Second, with limited traction, the car will have lower forces acting on it in pitch and roll. A softer setting will allow the car to move the same amount that it does in the dry with higher forces and stiffer settings. The alignment settings, especially camber, are designed to be optimum when the car rolls due to cornering forces; and the softer set-up allows the car to get to this range with lower lateral forces working on it. How much to soften your particular racecar is, again, something you have to find out by testing.

Pedal Changes

Take a close look at the pedals in your car. If the faces of the pedals, as they are in many racecars, are made of a smooth metal surface, they will be very slippery when wet. It's worth the effort to have rain pedals made, especially a custom brake pedal, with a face made of a rough surface such as expanded metal screen. Non-skid tape works in a pinch. Also, your racing shoes, with their thin leather soles, can get incredibly slippery when they get wet. To prevent your



Fig. 12-3. In wet conditions, drying the soles of your driving shoes makes them less likely to slip off the pedals.

feet from slipping off the pedals in the rain, dry your soles or try wearing deck shoes for better traction.

Take a look at the pedal heights as well. Normally, the relationship between brake pedal and throttle pedal height is set so that under hard braking, the throttle can be easily reached with the right side of your right foot in order to blip the throttle on the downshifts. When it's wet, you won't be pushing on the brake pedal as hard and you might find the throttle too low to do good blips. Always remember to re-adjust the brake-throttle relationship before racing in the wet. It will be much easier to spin on a mismatched downshift in the wet, so give yourself some help by getting the pedal relationship right.

Engine

Motors, especially their electrics, don't like water, so find out in testing whether you have a problem with too much water entering the engine compartment. Pay special attention to air intakes for the induction system. A ram air set-up which is terrific in the dry can drown a motor in the rain. An alternate air source that is clear of direct spray usually does the trick.

If you race a car with big horsepower, your biggest problem in the wet will be trying to keep the driving wheels from spinning. Take a close look at how much throttle pedal travel you normally have. If one inch of travel covers the range from minimum to 600 hp, it's going to be very difficult to modulate out of wheel-spin. By making up a special bellcrank or throttle pedal, you can greatly increase the amount of total pedal travel so that it's easier to make subtle changes in throttle position.

"When we built the Eagle Formula Ford, we had five or six positions on the throttle pedal where we could affix the throttle cable, so we could have the choice of a quick throttle for some racetracks and we'd be able to give it a long throttle arm for the wet so you could modulate it better. Gurney fit the same type of set-up to the Indy Cars so that when you went to road races you could have a longer throttle throw than you would on an oval."

—David Loring

Gearing

Some drivers feel that using a higher-than-normal gear makes it easier to avoid wheelspin in the wet. Another school of thought maintains that using a lower gear allows better throttle control and better straight-line acceleration. Ultimately, it's up to the individual driver—and the stopwatch—to decide which is the better course. We vote for using an RPM range that produces the most horsepower, as well as for controlling the wheelspin with your right foot.

Always Test First

In 1980, John Fergus, 1990 IMSA GTU Champion and former Pro S 2000 Champ, ran a Crossle S-2000 car, the first of its kind in the U.S. Its inaugural race was an SCCA National at Moroso Motorsports Park in Florida and, as is common with new cars, it arrived

"I've gone to using a higher gear in cornering in the wet to try and tame down the torque available and the wheelspin capability. I've done that and sometimes gone down a gear once I'm through the corner where you can get a little more grip and use the extra torque to shoot you down the straight."

—Terry Earwood

at the last minute with only one test day available to see what we had.

It rained on the test day, of course, and we were tempted to call it off but thought better of it. We went ahead and mounted the rain tires and rolled this absolutely new car out into the muddy paddock. John roared off onto the circuit and one lap later was back in. The car was undriveable in the wet, not because of a handling problem, but because of its aerodynamics. The shape of the nose bodywork was such that, at speed, it concentrated the spray of water off the nose directly into John's face. "Just like looking into a fire hose," was John's comment.

We fixed it with a jury-rigged spoiler and a wind-shield extension. Did I mention that you should test in the wet?

RAIN-DRIVING TECHNIQUES

Once you solve the vision problem and have got the car sorted out, the final adaptation to the rain is how and where you drive the car.

Find a New Line

In the dry, all our efforts in terms of line are geared toward finding the biggest radius possible for each corner in order to maximize cornering speed. In the rain, this changes. It's possible, especially if there's a deep puddle at the normal apex, that the cornering force of the car can go to zero. With zero cornering force, it doesn't matter how big you try to make the radius of the corner: the car is going straight.

In rain driving, the classic dry line becomes secondary to the effort to find cornering grip. Often the

"In the rain, there's no rules. There are some cases where you drive the 'traditional rain line,' some where the dry line works and others where you drive something in between. You just go out and look for traction. The guy who is comfortable in the car in the wet, has good car control, and goes and finds the traction is going to go the quickest."

—Jeremy Dale



Fig. 12-4. In the dry, the concentration is on carving the biggest radius through corners, while in the wet, the focus is on maintaining traction by avoiding areas, like this deep puddle at the apex, where the traction could go to zero.

dry line is the last place you want to be when it's pouring out. Since the line carries most of the racing traffic, the track is often polished smooth from rubber being laid down, filling the tiny irregularities in the track surface. Water just sits on this polished surface and easily forms a sheet between your tires and the road. To make matters worse, all of the spilled oil and antifreeze from previous races lies on the conventional line. Put the two together and the dry line can be practically undriveable.

Go off-line, however, and there is less spilled lubricant and the track surface is much more porous, giving the tires much better bite.

Another advantage of driving outside the dry line is that any banking in the corner is going to allow the water to run away from the outside of the turn down toward the inside. The outside edge of the corner will simply be drier than the inside and, given the choice, you want to drive on the driest pavement.

We did an experiment at Lime Rock, using our data-collection Formula Dodge, where we did back-to-back rain sessions on the dry line vs. driving around the outside rim of every corner. The average lap time around the outside was eight seconds per lap faster. Eight seconds!

The rim-shot method (see Fig. 12-6) worked in our test, but since wet weather is so unpredictable, there is no one right way to modify your line. The key is to continually experiment, trying to find a faster method. An indispensable tool for this experimentation is the tachometer. Just as you should in the dry, you check the tach reading at a consistent point at the corner exit to see if a change in approach yielded better corner-exit speed. As conditions change, as it rains harder, more lightly, or begins to dry, you continue to try different approaches and use the one that gives the best corner-exit speed.



Fig. 12-5. Track surfaces can become highly polished as the surface irregularities are worn down by constant use. Here, the dry line approaching the uphill at Lime Rock shimmers in the late-afternoon sun.

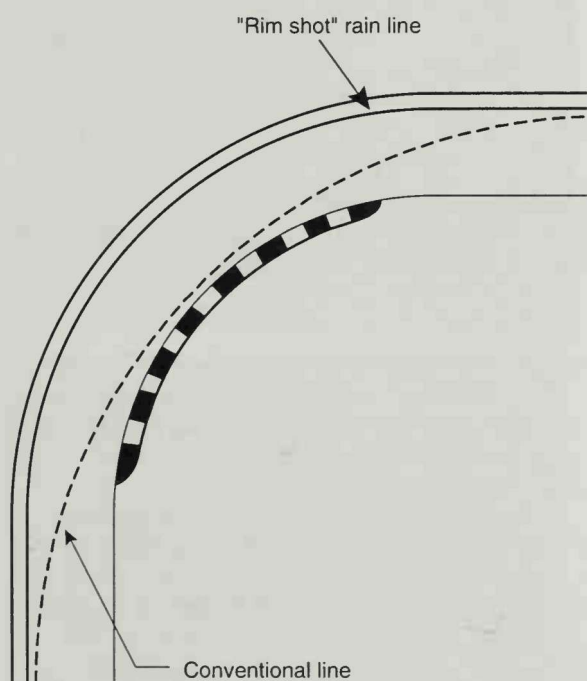


Fig. 12-6. The "rim shot" takes advantage of the superior grip available outside the polished dry line.

Your Line Options

There are two parallel concerns in trying to decide on the best rain line. One is that, as we covered above, you want to keep speed as high as you can while in the corner. You do this by finding the grippiest pavement, staying away from the slippery stuff down around the normal apex. In our Lime Rock experiment, the cornering force available on the dry line was in the neigh-

borhood of .6 Gs. On the outside, the G readings were in the vicinity of .85 Gs, some 42% better. By using the outside route, you started accelerating down the following straight from a higher speed and, given more total grip, you could use more throttle earlier than you could on the dry line.

The rim-shot technique works especially well in long, sweeping corners where you'll spend a lot of time cornering. The more time spent cornering, the longer you have that 42% grip advantage and the bigger the lap time advantage. Even though the radius of the dry line is bigger, the limited traction still means a loss in speed. In West Bend, a long sweeper at Lime Rock, the speed around the outside was 6 m.p.h. faster than the dry line. You also carry that 6 m.p.h. advantage onto the straight.

Another phenomenon that has bearing on your approach to rain lines is that tires lose more of their cornering ability in the wet than they lose either braking or accelerating grip. Relative to the dry, our tires lose about 36% of their dry weather traction in braking or accelerating. The cornering ability, however, falls 50%.

In light of these facts, you can decide to focus on the tires' strengths when it rains and try to minimize the time spent turning. You would do this by trying to square off the corners—that is, turn later at a slower speed and get the car pointing as straight as possible as early as possible, using aggressive acceleration early in the corner. It's like turning the racetrack into a series of dragstrips and braking zones, connected by low-speed turns where you get the car's angle changed over a short distance and time (see Figure 12-7).

This technique works really well on courses with tight short corners but not so well on racetracks with

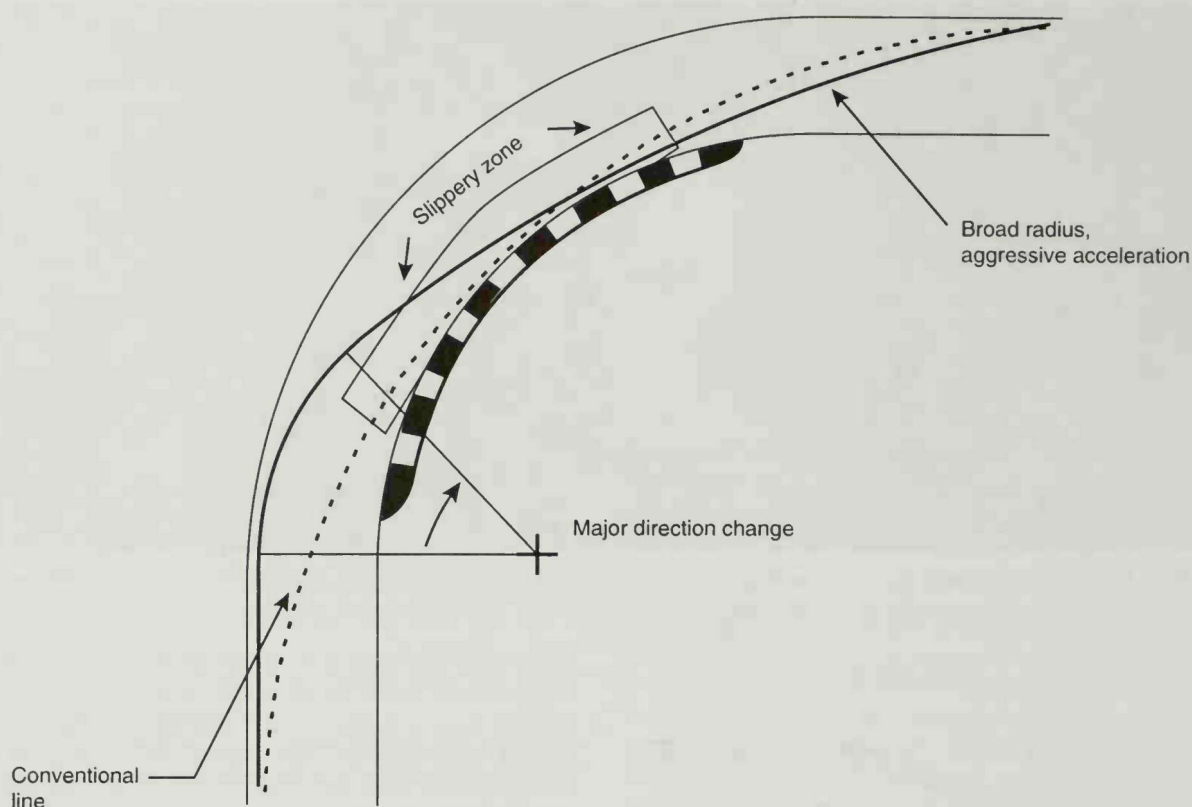


Fig. 12-7. In tight, short corners, one option is to do the bulk of the car's change of direction early in the turn, allowing a much straighter acceleration path at the corner exit. This does require passing over the slippery dry line and the throttle will have to be modulated to account for the changes in grip on this path.

long sweepers and hairpin turns, where the rim shot is generally more effective.

With either approach, you have to cross the dry line at some point, so expect that you may have to modulate the braking pressure on the way into the corner and breathe out of the throttle at the corner exit as you pass over the more slippery pavement.

The Makeup of the Circuit

Your choice of line and approach will vary from track to track, even corner to corner, based on the layout of the course and what the road surface is like. At Lime Rock, a circuit with three fast sweepers, two long hairpins and one banked medium-speed corner, the rim shot technique is the way to go, as you've seen. This is especially so because the track surface is all asphalt and polished from years of racing.

At a place like the Sebring test circuit, there are four short, slow corners where the track surface is made of porous, abrasive concrete. The concrete drains well and the water seeps into the surface pores; consequently, the apex area of these four turns isn't any worse than the rim as far as traction is concerned. In these four corners, you can square off the turn by using a late apex and get a good straight run off the

"I spun a car once in the rain at Watkins right on the pit straight. I hit a puddle with the right side of the car—I couldn't see well enough to realize where the water was deep. I'd have never hit it knowingly with one side of the car. The car did a 360 and kept going, and I didn't lose a half-second. I didn't have anything to do with the fact that I ended up going in the right direction—it just happened."

—Skip Barber

corner even though you're crossing the dry line. The other fast corners at Sebring have an asphalt surface and driving off-line, much like you do at Lime Rock, creates some grip advantage in these corners.

Straights

In dry weather, the path down the straight is obvious. You track out of the corner leading onto the straight, then use the straight to position your car for the entry of the next corner. In the rain, it's possible to get wheelspin down the straight, even in relatively low-



Fig. 12-8. Over time, ruts form in the braking zones, then fill with water when it rains. Braking is much better by keeping the tires on the dryer crown adjoining the ruts.

"At New Orleans, in the Lights car with Essex Racing, it rained and stopped, rained and stopped, and we changed tires so often in that race that I forgot what tires I was on—which ultimately led to our demise. We were leading with 20 minutes to go when I went off-line to pass a backmarker and, just as I went to the brakes, it dawned on me that he was on rain tires and I was on slicks. I slid right past him, right out of the lead and into the tire barrier.... I got caught out on that one just a bit."

—Jim Pace

powered cars, if you run on the wettest part of the track. Your approach to the straight is to find the driest path, and that may not be in the accustomed place. You certainly want to avoid hitting puddles, especially hitting them with only one side of the car. Hitting deep water at speed has a formidable braking effect. Ask anyone who has done a belly-flop off a bridge.

Hitting a puddle with one side of the car can spin it right out from under you. If there's no way around it, like when it extends from one verge of the track to the other, try to hit it square so that the car will skim over the top and arrive on the other side pointing the same way it started.

In the braking zone at the end of the straight, don't lock into the idea of using the same part of the racetrack as you do in the dry. Many corners have ruts worn into them at the approach to corners and these little hollows fill up with water, robbing you of braking traction. Often just straddling these ruts, placing your tires on the crown rather than the trough, will improve the car's braking potential.

Once the race has settled down and drivers find a workable line, it will tend to get drier unless you're

faced with a continual downpour. You have to keep in mind that when you pull off of this drier line to pass another driver in the braking zone for a corner, you'll be back on the really wet stuff. Outbraking in the wet can be dicey, especially in drying conditions where the off-line can be considerably more slippery than the path most drivers are on.

COMMON-SENSE TIPS

In the rain, little things can catch you out. With the limited traction, any of the mistakes that you normally make in your dry-weather racing are magnified. The mismatched downshift that you got away with in the dry can get the rear of the car sliding in the wet. Ditto for trailing throttle oversteer. You have to try to be as smooth as you can with all the controls.

Always look for drier or more abrasive asphalt. If the track surface looks glossy, it's probably slippery. If it's dull gray in appearance, there's probably some grip there. Look for patches in the racetrack surface. Some, like Mid Ohio's dreaded epoxy sealer, are like ice. Concrete patches, especially if they are freshly poured, have great grip.

A lot of your competitors will spin in the wet. Just going the distance without going off the road can allow you to finish well. If you do spin, try to be cool. We've seen so many people do a second spin in an effort to hurry back onto the track that you'd think it's required in the rules. If you think the track is slippery, grass raises "greasy" to another level of magnitude.

If you prepare for the wet, it really shouldn't be any more intimidating than racing on a sunny day. It has its own special requirements, but if you have good driving gear, a good car, and an experimental attitude, you can consistently outperform the many other drivers who just want to get it over with.